

207 Final Project Proposal

Section 5, Group 3

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Project GitHub repo: <https://github.com/cyrustaba/datasci-207-final-project>

Motivation:

Question: Can we accurately predict a track's genre or popularity score based purely on its audio spectrogram image?

We want to investigate this question because it allows us to treat an audio problem as an image classification task by utilizing spectrograms. We are essentially testing whether a spectrogram alone can capture enough information to replace the metadata that music platforms typically rely on for genre tagging and popularity scoring. This is also a great opportunity for us to apply the image classification methods we will cover in class to a domain outside the typical use case, while working with data in a near-raw form.

Data:

Our dataset is sourced from the Free Music Archive (FMA), a public collection of Creative Commons-licensed audio maintained in the following GitHub repository: <https://github.com/mdeff/fma>. The repository provides the raw audio as .mp3 files, packaged into several subsets of increasing size and complexity, ranging from a curated 8,000-track sample with 8 balanced genres and clips trimmed to 30 seconds, up to the full collection of 106,574 untrimmed tracks spanning 161 unbalanced genres (roughly 879 GiB).

For this project, we plan to use the 8,000-track balanced subset (FMA-small) as our primary dataset, since its manageable size and balanced genre distribution make it well-suited for initial model development. If time and computational resources allow, we may scale up to a larger subset later in the project. We have not yet finalized our outcome variable and are currently weighing two candidates: genre classification and popularity prediction. Since genre classification is already included in the initial dataset and does not need any additional transformation, we will start with genre classification and consider an additional estimation of popularity if time allows or in future projects.

Related work:

Defferrard et al. (2017), "FMA: A Dataset for Music Analysis".

<https://arxiv.org/abs/1612.01840>

Describes the dataset's construction, and how to use it.

Choi et al. (2016), "Explaining Deep Convolutional Neural Networks on Music Classification".

<https://awesomepapers.io/speech-audio/papers/choi2016explaining>

Describes learning and prediction for Music Classification using CNNs applied to spectrograms.

Tudor, Alex, et al. (2020), "The Machine Learning Behind Hum to Search." Google Research.

<https://research.google/blog/the-machine-learning-behind-hum-to-search/>

Describes how Google's Hum to Search learns melody embeddings from spectrograms.

Methodology (optional):

Using the librosa package in Python, we will convert waveforms to Mel spectrograms, which we will then feed as images into our imaging model.

We will also leverage SHAP to explain why the model predicted a given popularity score, as it can scan the model's predictions and identify which regions of the spectrogram image most influenced the outcome.